ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)
JAIPUR CENTER
Academic Session : 2021-22

SRGBT

for
JEE (MAIN+ADVANCED) 2022
ENTHUSIAST COURSE

Date : 10-10-2021 | Time : 3.00 Hr | CLASS-XII

Student's Form No. : ..... Batch : .....

Student's Name : .....

INSTRUCTIONS

1. PART-1 : Physics : (Q. No. 1 to 6) : Single Correct Type Questions : Marking Scheme (+3, –1)
   (Q. No. 7 to 14) : One or More Correct Type Questions : Marking Scheme (+4, –2)
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 15 to 18) : Comprehension based Questions : Marking Scheme (+3, –1)
2. PART-2 : Chemistry : (Q. No. 19 to 24) : Single Correct Type Questions : Marking Scheme (+3, –1)
   (Q. No. 25 to 32) : One or More Correct Type Questions : Marking Scheme (+4,–2)
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 33 to 36) : Comprehension based Questions : Marking Scheme (+3, –1)
3. PART-3 : Mathematics : (Q. No. 37 to 42) : Single Correct Type Questions : Marking Scheme (+3, –1)
   (Q. No. 43 to 50) : One or More Correct Type Questions : Marking Scheme (+4,–2)
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 51 to 54) : Comprehension based Questions : Marking Scheme (+3, –1)

PART-1 : PHYSICS

SINGLE CORRECT TYPE (+3, –1)

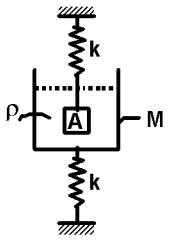
1. The system shown in the adjacent figure is in equilibrium. The mass of the container with liquid is M, density of liquid in the container is ρ and volume of the block is V. If the container is now displaced downwards through a distance x_0 and released such that the block remains well inside the liquid, then during subsequent motion

(A) time period of SHM of the container will be $2\pi\sqrt{\frac{M}{k}}$

(B) time period of SHM of the container will be $2\pi\sqrt{\frac{(M+\rho v)}{k}}$

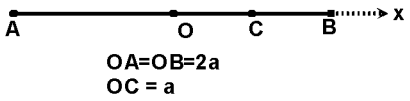
(C) amplitude of SHM of the container will be greater than x_0

(D) container will not perform SHM



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2. A particle of mass m is performing simple harmonic motion along line AB with amplitude 2a with centre of oscillation as O. At t = 0 particle is at point C (OC = a) and is moving towards B with velocity v = $a\sqrt{3}$ m/s. The equation of motion can be given by



(A) x = 2a (sin t + $\sqrt{3}$ cos t)

(B) x = 2a ($\sqrt{3}$ sin t + cos t)

(C) x = a (sin t + $\sqrt{3}$ cos t)

(D) x = a ($\sqrt{3}$ sin t + cos t)

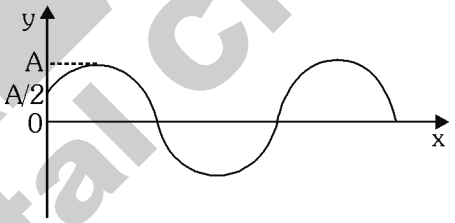
3. 200 students are talking in an examination in a room, and the sounds of pens scratching on paper, sighs, groans, and muttered imprecations has created a more or less continuous sound level of this noise of 60dB. Assuming each student contributes equally to this noise and nothing else changes or adds to it, what will the sound level in the room be when only 50 students are left ? ($\log_{10} 2 = 0.3$)

(A) 50 dB (B) 15 dB (C) 66 dB (D) 54 dB

4. A sonometer wire vibrates with a frequency f. It is replaced by another wire of three times the diameter if tension and other parameters remain the same, the frequency of vibration of the wire will be-

(A) $\frac{f}{9}$ (B) $\frac{f}{3}$ (C) 3f (D) 9f

5. Here snap shot of a progressive wave at $t = \frac{T}{6}$ is given with time period = T. Then the equation of the wave will be if wave is going in positive x-direction and if wave is going in negative x-direction respectively. $\left(\text{Here, } T = \frac{2\pi}{\Omega}\right)$ (symbol have their usual meaning)



(A) $y_1 = A\sin\left(kx - \omega t + \frac{\pi}{2}\right)$, $y_2 = A\sin\left(kx + \omega t - \frac{\pi}{6}\right)$

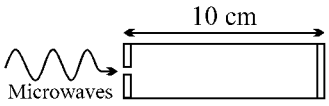
(B) y_1 = A cos (kx – ωt), y_2 = A sin (kx + ωt)

(C) y_1 = A sin (kx – ωt), $y_2 = A\sin\left(kx + \omega t - \frac{\pi}{6}\right)$

(D) $y_1 = A\sin\left(kx - \omega t + \frac{\pi}{2}\right)$, $y_2 = A\sin\left(kx + \omega t + \frac{\pi}{6}\right)$

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6. You have a microwave generator that can produce microwaves at any frequency between 1GHz and 10 GHz. The microwave radiation enters a 10-cm-long cylinder with reflective end caps, as shown in the Figure. What frequency of the microwave generator, in GHz, will produce the lowest-order standing wave with an antinode (maximum) in the center of the cavity? Note that with reflectors at both ends, the electromagnetic standing wave acts just like the standing wave on a string that is tied at both ends. Take velocity of waves as 3×10^8 m/s.



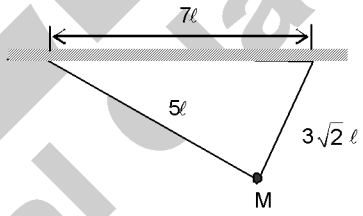
(A) 4.5 (B) 3.0 (C) 1.5 (D) 7.5

ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

7. A particle is suspended by two ideal strings as shown in the figure. Now mass m is given a small displacement perpendicular to the plane of triangle formed. Choose the correct statement(s).



(A) The period of oscillation of the system is $2\pi\sqrt{\frac{3\sqrt{3}\ell}{g}}$.

(B) The period of oscillation of the system is $2\pi\sqrt{\frac{3\ell}{g}}$.

(C) The period of oscillation of the system is independent of M.

(D) If the distance between the suspension points was kept constant and the lengths of the strings were quadrupled then the period of the system will double.

8. Two particles A and B are performing SHM along x and y-axis respectively with equal amplitude and frequency of 2 cm and 1 Hz respectively. Equilibrium positions of the particles A and B are at the co-ordinates (3, 0) and (0, 4) respectively. At t = 0, B is at its equilibrium position and moving towards the origin, while A is nearest to the origin and moving away from the origin. Choose the correct option(s) :

(A) The distance between A and B is 3 cm at t = t_0 sec. It will again be 3 cm at t = (t_0 + 1) sec.

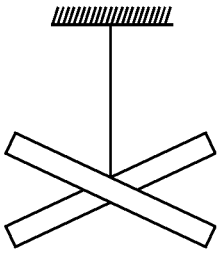
(B) The distance between A and B can never be 5 cm.

(C) The maximum magnitude of force acting on each particle during the motion is same.

(D) Kinetic energy of the system (A and B) is conserved

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9. A torsional pendulum consists of two identical rods each of mass M & length ℓ joined perpendicular to each other in horizontal plane and is hanged through the wire having torsional constant C. The system performs simple harmonic motion $\theta = A\sin(\omega t + \phi)$ such that initially angular displacement of the system is equal to half of amplitude and its angular velocity is ω_0 and is decreasing, then :-



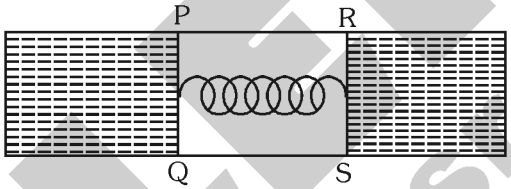
(A) Time period of pendulum is $T = 2\pi\sqrt{\frac{M\ell^2}{6C}}$

(B) Amplitude of oscillation is $\frac{\omega_0\ell}{3}\sqrt{\frac{2M}{C}}$

(C) Initial phase of SHM is $\frac{\pi}{6}$ or $\frac{7\pi}{6}$

(D) Energy stored in wire at $t = 0$ is $\frac{M\ell^2\omega_0^2}{36}$

10. Two identical straight wires PQ and RS each of mass m and length ℓ can move smoothly on a fixed rectangular frame. Two thin films of a liquid of surface tension T are formed between each wire and the frame. The two wires are connected by a massless spring of stiffness k and intially in a natural length position and released then choose the correct option(s).



(A) Maximum elongation of spring, $\Delta x_m = \frac{2T\ell}{k}$

(B) Maximum elongation of spring, $\Delta x_m = \frac{4T\ell}{k}$

(C) Each wire executes SHM with the time period, $T_0 = 2\pi\sqrt{\frac{m}{2k}}$

(D) Each wire executes SHM with the time period, $T_0 = 2\pi\sqrt{\frac{2m}{k}}$

11. The first overtone of an open organ pipe (of length L_0) beats with the first overtone of a closed organ Pipe (of length L_c) with a beat frequency of 10 Hz. The fundamental frequency of the closed pipe is 110 Hz. If the speed of sound is 330 m/s. Then

(A) $L_c = 0.75$ m

(B) $L_o = \left(\frac{33}{34}\right)$m or $\left(\frac{33}{35}\right)$m

(C) $L_o = \left(\frac{33}{32}\right)$m or 1 m

(D) $L_o = \left(\frac{33}{32}\right)$m or $\left(\frac{33}{34}\right)$m

12. A vibrator makes 150 cm of a string to vibrate in 6 loops in the longitudinal arrangement when it is stretched by 15 g. Then The entire length of the string is then weighed and is found to weigh 500 milligram.

(A) frequency of the vibrator is 84 Hz

(B) frequency of the vibrator is 42.4 Hz.

(C) distance between two nodes is 25 cm

(D) distance between two nodes is 33 cm

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13. The wave function for the wave pulse is $y(x, t) = \frac{0.1C^3}{C^2 + (x - vt)^2}$, with C = 4 cm at x = 0 the displacement y(x, t) is observed to decrease from its maximum value to half of its maximum value in time $t = 2 \times 10^{-3}$ s.

(A) y(x, t) represents the motion of travelling pulse moving along positive X direction

(B) y(x, t) does not represent the motion of a travelling pulse

(C) Speed of wave pulse is 20 m/s

(D) Speed of wave pulse is 10 m/s

14. There are three sound sources of frequencies 600Hz, 603Hz and 609Hz sounded simultaneously. Initially they are in same phase. A detector is placed close to them. Let I_0 be the intensity of each sound source. If detector is able to detect intensity nearly equal to

(A) $3I_0$, the frequency detected by the detector is 9Hz

(B) $3I_0$, the frequency detected by the detector is 6Hz

(C) $9I_0$, the frequency detected by the detector is 3Hz

(D) $9I_0$, the frequency detected by the detector is 6Hz

COMPREHENSION TYPE (+3, –1)

Paragraph-1 for Q. Nos. (15 & 16)

A string 120 cm in length sustains a standing wave with the points of the string at which the displacement amplitude is equal to 3.5 mm being separated by 15.0 cm. Find

15. The maximum displacement amplitude

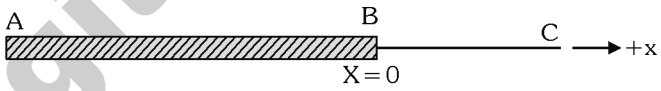
(A) $\sqrt{2}$ mm (B) 2.95 mm (C) 4.95 mm (D) 6.95 mm

16. To which overtone do these oscillations correspond ?

(A) 4 (B) 3 (C) 5 (D) 6

Paragraph-2 for Q. Nos. (17 & 18)

In the figure shown a sinusodial wave is generated at the end 'A'. The wave travels along positive x-axis and during its motion it encounter another string BC at the junction 'B' at x = 0. The density of strings AB and BC are ρ and 9ρ respectively and their radii of cross sections are 2r and r. The wave function, amplitude and wavelength of incident wave are respectively y_i, A_i and λ_i. Similarly for reflected and transmitted wave these parameters are y_r, A_r, λ_r and y_t, A_t, λ_t respectively.



17. The ratio of wavelengths λ_r to λ_t (i.e. $\lambda_r : \lambda_t$) will be

(A) 1 : 1 (B) 3 : 2 (C) 2 : 3 (D) None of these

18. The ratio of amplitudes A_r to A_t is (i.e. $A_r : A_t$) will be

(A) 1 : 1 (B) 1 : 4 (C) 4 : 1 (D) None of these

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PART-2 : CHEMISTRY

SINGLE CORRECT TYPE (+3, –1)

19. The solubility of ethylamine in a solution with pH = 8.5 is (K_{sp} of ethylamine = 1.78×10^{-10})
(Given: $\sqrt{10} = 3.16$ **)**
(A) 5.6×10^{-3} M (B) 3.8×10^{-4} M (C) 5.6×10^{-5} M (D) 3.16×10^{-6} M

20. A 250 ml flask containing $NO_{(g)}$ at 0.42 atm is connected to a 100 ml flask containing oxygen gas at 0.84 atm by means of a stop cock at 350 K. The gases are mixed by opening the stop cock where the following equilibrium is established.

$$2NO(g) + O_2(g) \longrightarrow 2NO_2(g) \rightleftharpoons N_2O_4(g)$$

The first reaction is complete while the second is at equilibrium. If the total pressure is 0.365 atm, K_p for dimersation of NO_2 is
(A) 10 atm^{-1} (B) 0.4 atm^{-2} (C) 0.16 atm^{-2} (D) 0.4 atm

21. In the excess of NH_3 (aq.), Cu^{2+} ions forms a deep blue complex ion $[Cu(NH_3)_4]^{2+}$ which has $K_f = 10^{11}$ the concentration of Cu^{2+} ion in solution prepared by adding 5×10^{-3} mole of $CuSO_4$ to 0.5 L of 0.4 M NH_3 (aq) is

(A) $\frac{5\times10^{-5}}{(36)^4}$ (B) $\frac{10^{-6}}{(36)^4}$ (C) $\frac{10^{-5}}{(36)^4}$ (D) $\frac{10^{-2}}{(39)^4}$

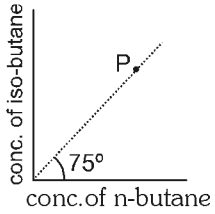
22. Equilibrium constants for the following reactions are given **(in atm)** for the reactions at 0°C. Select the option mentioning correct order of **True (T)** or **False (F)** statement.

A. $6H_2O_{(s)} \rightleftharpoons A.2H_2O_{(s)} + 4H_2O_{(g)}$ $K_P = 1.6 \times 10^{-11}$
B. $12H_2O_{(s)} \rightleftharpoons B.7H_2O_{(s)} + 5H_2O_{(g)}$ $K_P = 2.43 \times 10^{-13}$
C. $10H_2O_{(s)} \rightleftharpoons C_{(s)} + 10H_2O_{(g)}$ $K_P = 10^{-30}$
Aqueous tension of H_2O at 0°C is given as 0.76 torr
(I) The most effective drying agent will be $C_{(s)}$ out of $C_{(s)}$, $B.7H_2O_{(s)}$ & $A.2H_2O_{(s)}$
(II) At 0°C $A.6H_2O_{(s)}$ & $B.12H_2O_{(s)}$ will always be efflorescent.
(III) If R.H is less than 100% in a chamber at 0°C then none of the substance can act as deliquescent.
(A) TTT (B) FTF (C) TTF (D) TFF

23. The indicator constant for an acidic indicator, HIn, is 5×10^{-6} M. This indicator appears only in the colour of acidic form when $\frac{[In^-]}{[HIn]} \le \frac{1}{20}$ and it appears only in the colour of basic form when $\frac{[HIn]}{[In^-]} \le \frac{1}{40}$. The pH range of indicator is
(A) 4.3 – 6.3 (B) 4.0 – 6.6 (C) 4.0 – 6.9 (D) 3.7 – 6.6

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24. For the equilibrium $CH_3 - CH_2 - CH_2 - CH_3(g) \rightleftharpoons CH_3-\underset{\displaystyle H}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-CH_3(g)$ equilibrium constant is found to be 1.732 at 298 K. Now if in a vessel at 298 K, a mixture of these two gases be taken as represented by the point P in the figure, predict what will happen :



(A) Immediately, above equilibrium will be set up

(B) Above reaction will be in the forward direction till it attains equilibrium

(C) Above reaction will go in the backward direction till it attains equilibrium

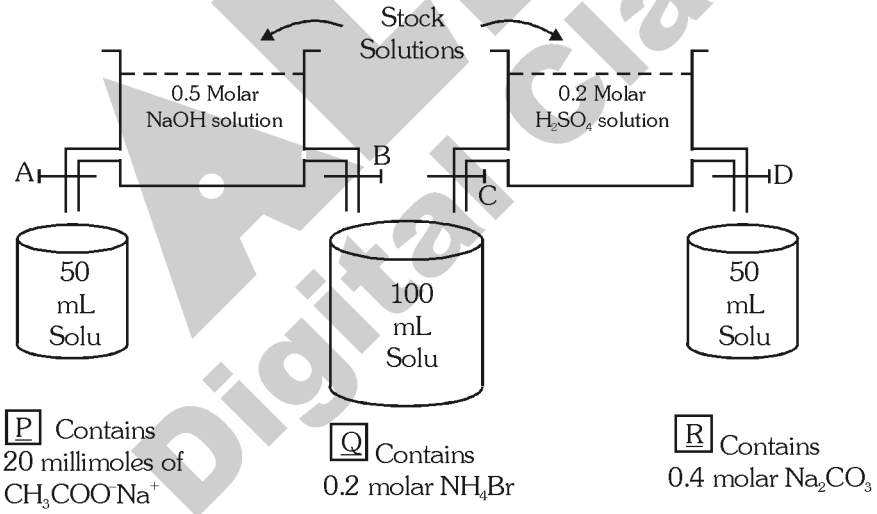
(D) Nothing can be said

ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
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Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

25.



If valves are opened sequentially (A to D) & each valve lets out 20 mL of stock solution, then select correct statements (Temp = 25°C)

(A) After opening (A), buffer is created in beaker "P".

(B) After opening (B), buffer is created beaker "Q".

(C) After opening (C), contents in beaker Q can still act as buffer even though solutions pH decreases.

(D) After opening (D), solution of pH < 7 is created in beaker "R".

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26. If two gases AB_2 and B_2C are mixed, following equilibria are readily established :

$$AB_2(g) + B_2C(g) \rightleftharpoons AB_3(g) + BC(g);$$

$$BC(g) + B_2C(g) \rightleftharpoons B_3C_2(g)$$

If the reaction is started only with AB_2 with B_2C, then which of the following is necessarily true at equilibrium ?

(A) $[AB_3]_{eq} = [BC]_{eq}$ (B) $[AB_2]_{eq} > [B_2C]_{eq}$ (C) $[AB_3]_{eq} > [B_3C_2]_{eq}$ (D) $[AB_3]_{eq} > [BC]_{eq}$

27. A solution contains 0.05 M each of NaCl and Na_2CrO_4. Solid $AgNO_3$ is gradually added to it. Which of the following facts is/are incorrect?

Given: $K_{sp}(AgCl) = 1.7 \times 10^{-10}\ M^2$ and $K_{sp}(Ag_2CrO_4) = 1.9 \times 10^{-12}\ M^3$.

(A) Cl^- ions are precipitated first

(B) CrO_4^{2-} ions are precipitated first

(C) Both Cl^- and CrO_4^{2-} ions are precipitated simultaneously

(D) The second ion starts precipitating when half of the first ion is precipitated

28. $CN^-_{(aq.)} + H_2O \rightleftharpoons HCN_{(aq.)} + OH^-_{(aq.)}$

Addition of which of the following reagent can make the reaction proceed towards forward direction.

(A) H_2O (B) HCl (C) NaBr (D) KOH

29. 1 M solution of a dibasic acid H_2A $\left(K_{a_1} = 10^{-6} \text{ and } K_{a_2} = 10^{-10}\right)$ is titrated against 1 M NaOH solution. Choose the **correct** statements regarding the relative concentration of H_2A, HA^- and A^{2-} at various pH during titration.

(A) At pH = 3 ; $[H_2A]$ is maximum. (B) At pH = 6.5 ; $[HA^-]$ is maximum.

(C) At pH = 10 ; $[H_2A]$ is minimum. (D) At pH = 8 ; $[A^{2-}]$ is maximum.

30. For the reaction $N_2O_5(g) \longrightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$

$N_2O_5(g)$ decomposed at a constant volume and temperature. Initial pressure of N_2O_5 is 600 mm of Hg and total pressure at equilibrium is 960 mm of Hg. Assuming ideal behavior of gases, Select the correct option(s).

(A) Involved reaction is intramolecular redox reaction

(B) The mole fraction of NO_2 in the equilibrium mixture is 0.5

(C) The equilibrium constant K_P for the above reaction is 540 mm of Hg

(D) For the above reaction $K_C = K_P (RT)^3$

31. Solubility product of AgCl (s) in water at 25°C is $12 \times 10^{-10}\ M^2$ then select the correct options.

(A) Solubility of AgCl in presence of 10^{-5} M $AgNO_3$ is 3×10^{-5} moles/L.

(B) Solubility of AgCl in presence of excess of KCl will increase.

(C) Solubility of AgCl in presence of 2M NH_3 is greater than that in pure water.

(D) Solubility of AgCl remains unaffected on adding any amount of NaBr because there is no common ion effect.

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32. Ammonia under a pressure of 20 atm at 127°C is heated to 327°C in a closed rigid vessel. Under these conditions, ammonia is partially decomposed to N_2 and H_2 according to the reaction

$$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$$

After decomposition, the pressure inceases to 50 atm. Which of the following statement(s) is/are **correct** at 327°C?

(A) The degree of dissociation of NH_3 is 2/3

(B) The K_p of the reaction is $2.7 \times 10^3 atm^2$.

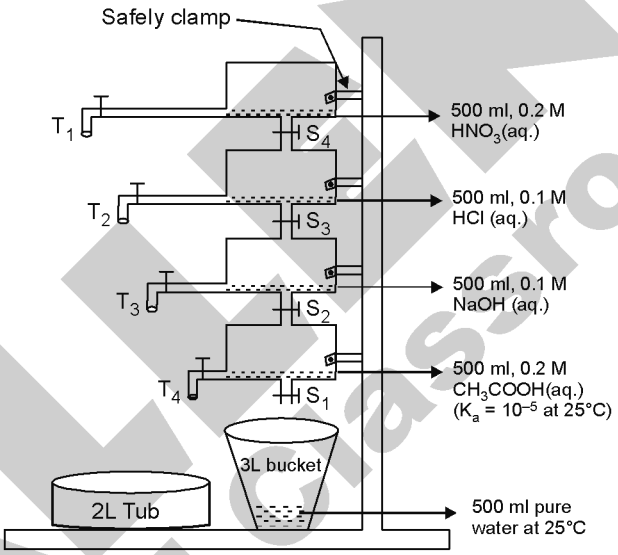
(C) The partial pressures of both, N_2 and NH_3 gas at equlibrium is 10 atm.

(D) The pressure of NH_3 at equilibrium is less than that of N_2.

COMPREHENSION TYPE (+3, –1)

Paragraph-1 for Q. Nos. (33 & 34)

Consider the following experimental set-up with stop cocks: S_1, S_2, S_3, S_4 and taps : T_1, T_2, T_3, T_4



33. In which of the following cases will the solution in tub have a pH different from others:

(A) T_1 and T_4 are opened

(B) only T_2 is opened

(C) T_1, T_2, T_3 and T_4 are all opened

(D) T_1 is opened and the content of bucket is poured in the tub.

34. A pH meter is immersed in the bucket of the original set up shown. It shows a reading of 7. Now S_1 is the first opened and all content is allowed to drain out into the bucket. pH of the solution in the bucket is measured again. Similarly S_2, S_3 and S_4 are sequentially opened and the pH of the solution in the bucket is measured each time after complete drainage. Opening which stop cock would cause maximum pH change (increase/decrease) compared to previous pH in bucket.

(A) S_1 (B) S_2 (C) S_3 (D) S_4

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Paragraph-2 for Q. Nos. (35 & 36)

Following reaction is at equilibrium in basic medium at 300 K.

I_2 (s) $\rightleftharpoons$ I^- (aq.) + IO^-_3 (aq.)

Equilibrium concentration at 300 K are , [I^-] = 0.1 M, [IO^-_3] = 0.1 M

Given ΔG°_f (I^-, aq) = – 50 kJ/ mol

ΔG°_f (IO^-_3, aq) = – 123.5 kJ/ mol

ΔG°_f (H_2O, *l*) = – 233 kJ/ mol

ΔG°_f (OH^- , aq) = – 150 kJ/ mol

$R = \frac{25}{3}$ J mol^{-1} K^{-1} $\log_{10} e = 2.3$

35. Value of equilibrium constant is

(A) 10^{25} (B) 10^{35} (C) 10^{30} (D) 10^{27}

36. pH value at equilibrium is

(A) 8 (B) 6 (C) 9 (D) 5

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PART-3 : MATHEMATICS

SINGLE CORRECT TYPE (+3, –1)

37. Let $P(z) = z^3 + az^2 + bz + c$, where a, b, c, $z \in R$. There exist a complex number ω such that the three roots of $P(z)$ are $2\omega - 4$, $\omega + 9i$, $\omega + 3i$, where $i^2 = -1$. Then value of $|a + b + c|$ is equal to :

(A) 121 (B) 136 (C) 79 (D) does not exist

38. The polynomial $p(x)=(1+x+x^2+.....+x^{17})^2-x^{17}$ has 34 complex roots of the form

$z_k = r_k\left[\cos(2\pi a_k)+i\sin(2\pi a_k)\right]$, k = 1, 2, 3.... 34 with $0 < a_1 \le a_2 \le a_3 \le ..... \le a_{34} < 1$ and $r_k > 0$, then value of $a_1+a_2+a_3+a_4+a_5$ is equal to :

(A) $\frac{1}{19}$ (B) $\frac{3}{19}$ (C) $\frac{159}{323}$ (D) $\frac{482}{591}$

39. Complex numbers a, b and c are zeros of $p(z) = z^3 + qz + r$, where q, $r \in R$ and $|a|^2 + |b|^2 + |c|^2 = 250$. If the points corresponding to a, b and c are in the complex plane are the vertices of right angled triangle with hypotenus 'h' then $\left(\frac{h^2}{125}\right)$ is equal to :

(A) 1 (B) 2 (C) 3 (D) 4

40. Coefficient of x^2 in $\underset{100 \text{ times}}{\left(.....(x(x-2)^2-2)^2-.....-2\right)^2}$ is equal to :

(A) $\frac{4^{199}-4^{99}}{3}$ (B) $\frac{4^{201}-4^{100}}{3}$ (C) $\frac{4^{99}-4^{50}}{3}$ (D) None of these

41. If $\sum_{r=1}^{2008}(2009-r)\cos\left(\frac{2\pi r}{2009}\right)=-\frac{n}{2}$, then n is equal to :

(A) 2008 (B) 2009 (C) 2016 (D) 2018

42. The coefficients of x^n in the polynomial $\left(x+{}^{2n+1}C_0\right)\left(x+{}^{2n+1}C_1\right)\left(x+{}^{2n+1}C_2\right).....\left(x+{}^{2n+1}C_n\right)$:

(A) 2^{n+1} (B) $2^{2n+1}-1$ (C) 2^{2n} (D) None of these

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ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

● Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

43. Let z_1 and z_2 are two complex numbers such that $z_1 + z_2 = z_1^2 + z_2^2 = \frac{2i}{\sqrt{3}}$ (where $i = \sqrt{-1}$) then :

(A) z_1 and z_2 lies in first quadrant
(B) z_1 and z_2 lies in second quadrant
(C) z_1 and z_2 lies in third quadrant
(D) z_1 and z_2 lies in different quadrant

44. If $S_n = 2(\omega+1)(\omega^2+1) + 3(2\omega+1)(2\omega^2+1) + 4(3\omega+1)(3\omega^2+1) + ...$ upto n terms then the correct statement(s) is are : (where $n \geq 3$ and ω is non real cube root of unity)

(A) S_n can not be a prime number
(B) unit digit of S_{24} is 4
(C) S_n is always divisible by n
(D) S_n is an increasingh sequence

45. For two non-zero complex numbers A and B if $A + \frac{1}{B} = \bar{A}$ and $\frac{1}{A} + B = \bar{B}$, then :

(A) A and B are purely imaginary
(B) A and B are purely real
(C) $AB = -\frac{1}{2}$
(D) $AB = \frac{1}{2}$

46. If 1, ω, ω^2 $.....\omega^{n-1}$ are the nth roots of unity, then the value of $(4-\omega)(4-\omega^2).....(4-\omega^{n-1})$ is equal to :

(A) $\frac{2^{2n}-1}{3}$
(B) can not be prime number
(C) ${}^nC_1 + {}^nC_2 3 + ... + {}^nC_n 3^{n-1}$
(D) $\frac{6^n+1}{5}$

47. Let z_1 and z_2 are two distinct non-real complex numbers such that $z_1 + \frac{1}{z_2} \in R$ and $z_2 + \frac{1}{z_1} \in R$, then value of $|z_1 z_2|$ may

(A) A prime number (B) A natural number (C) Positive integer (D) Negative integer

48. If $(1 + x)^n = c_0 + c_1 x + c_2 x^2 + ... + c_n x^n$, where n is a positive integer, then

(A) $c_0 - c_2 + c_4 - ..... = 2^{n/2} \cos\left(\frac{n\pi}{4}\right)$
(B) $c_1 - c_3 + c_5 - ..... = 2^{n/2} \sin\frac{n\pi}{4}$
(C) $c_0 + c_4 + c_8 + ... = 2^{n-2} + 2^{\frac{n-2}{2}} \cos\left(\frac{n\pi}{4}\right)$
(D) $c_0 + c_3 + c_6 + ... = \frac{1}{3}\left(2^n + 2\cos\frac{n\pi}{3}\right)$

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49. Let $a_n = \frac{(1000)^n}{n!}$ for $n \in N$. Then a_n is greatest, when

(A) n = 998 (B) n = 999 (C) n = 1000 (D) n = 1001

50. If $(1 + 2x + 3x^2)^{10} = a_0 + a_1 + a_2x^2 + \ldots\ldots + a_{20}x^{20}$ then

(A) $a_1 = 20$ (B) $a_2 = 210$ (C) $a_4 = 8085$ (D) $a_{20} = 2^2 \cdot 3^7 \cdot 7$

COMPREHENSION TYPE (+3, –1)

Paragraph-1 for Q. Nos. (51 & 52)

Let c_1 and c_2 be the two curves on the complex plane defined as

$c_1 : z + \bar{z} = 2|z-1|$ and $c_2 : \arg(z+1+i) = \alpha$, where $\alpha \in (0, \pi)$,

such that curves c_1 and c_2 touch each other at $p(z_1)$.

51. A particle starts from a point $P(z_0)$. It moves horizontally away from origin by 2 units and then vertically away from origin by 3 units to reach at a point $Q(z_1)$. Then $|z_1|$ is equal to :

(A) 3 (B) 5 (C) 7 (D) 9

52. If $P(z_0)$ is rotated about origin through an angle 2α in clockwise sense, then the area bounded by c_1, and the line joining $P(z_0)$ and $Q(z'_0)$ is, (where $Q(z'_0)$) is the new position of $P(z_0)$ after rotation :

(A) $\frac{5}{6}$ sq. unit (B) 1 sq. unit (C) 2 sq. unit (D) $\frac{2}{3}$ sq. unit

Paragraph-2 for Q. Nos. (53 & 54)

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots\infty$$

$$f(x) = f(0) + f'(0)x + f''(0)\frac{x^2}{2!} + \ldots\ldots\infty$$

53. If $\frac{e^x}{1-x} = B_0 + B_1x + B_2x^2 + \ldots\ldots + B_nx^n + \ldots\ldots$ then the value of $B_n - B_{n-1}$ is equal to :

(A) n! (B) (n + 1)! (C) $\frac{1}{n!}$ (D) $\frac{1}{(n+1)!}$

54. Sum of the series $\left(\frac{2}{3!} + \frac{4}{5!} + \frac{6}{7!} + \ldots\ldots\infty\right)$

(A) e^2 (B) e^{-1} (C) e^{-2} (D) None of these

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ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

SRGBT-12

ENTHUSIAST COURSE DATE : 10-10-2021

ANSWER KEY

PART-1 : PHYSICS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	D	B	A	C	BC	AC	ABCD	BC
Que.	11	12	13	14	15	16	17	18		
Ans.	AC	BC	AC	AC	C	B	B	B		

PART-2 : CHEMISTRY

Que.	19	20	21	22	23	24	25	26	27	28
Ans.	C	B	C	A	C	C	BC	CD	BCD	AB
Que.	29	30	31	32	33	34	35	36		
Ans.	ABC	AB	ABC	ABC	C	A	C	A		

PART-3 : MATHEMATICS

Que.	37	38	39	40	41	42	43	44	45	46
Ans.	B	C	C	A	B	C	AB	ABD	AC	AC
Que.	47	48	49	50	51	52	53	54		
Ans.	BC	ABCD	BC	ABC	B	D	C	B		